

Algebraic and geometric methods in the theory of subspace codes

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Abstract

Subspace codes have emerged as a central object in network coding, where information is transmitted via subspaces of a finite vector space rather than individual vectors. Their study naturally brings together ideas from algebra, geometry, and combinatorics. In this talk, we examine the role of algebraic and geometric structures in the construction and classification of subspace codes, with particular emphasis on cyclic subspace codes and *Sidon spaces*. The latter provide a powerful framework for constructing subspace codes with optimal minimum distance, due to their extremal behavior with respect to the multiplicative structure of the ambient field. We present explicit constructions of subspace codes derived from Sidon spaces and show how their algebraic properties yield optimal or near-optimal parameters. On the classification side, we explore linear analogues of results from additive combinatorics, such as sumset estimates and inverse theorems, in the setting of finite field extensions. These tools offer structural insight into subspaces with constrained intersection patterns, leading to classification results for specific families of subspace codes.

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